

# Statistics Hybrid Practice Set – S4 (L1)

## Topics

- Population
- Sample
- Mean
- Median
- Mode
- **Range (New)**

## Instructions

- Solve on **paper first**.
  - Then solve the **same question in Excel**.
  - Write the Excel formula you used.
  - No calculator.
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## Q1. Identify Population and Sample

A hospital has **4,500 patients**. A researcher selects **180 patients** for a health survey.

## Paper Task

- Population = ?
- Sample = ?

## Excel Task

| Total Patients | Selected Patients |
|----------------|-------------------|
| 4500           | 180               |

(No formula required.)

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**Q2. Calculate the Mean**

| Value |
|-------|
| 9     |
| 13    |
| 17    |
| 21    |
| 25    |

**Paper Task**

- Calculate the Mean.

**Excel Task**

- Calculate the Mean.
  - Write the Excel formula.
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### Q3. Find the Median

| Value |
|-------|
| 22    |
| 10    |
| 14    |
| 18    |
| 26    |

#### Paper Task

- Find the Median.

#### Excel Task

- Calculate the Median.
  - Write the Excel formula.
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#### Q4. Find the Mode

| Value |
|-------|
| 6     |
| 8     |
| 8     |
| 10    |
| 12    |
| 14    |

#### Paper Task

- Find the Mode.

#### Excel Task

- Calculate the Mode.
  - Write the Excel formula.
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### Q5. Choose the Best Measure

Monthly Sales (Units)

| Value |
|-------|
| 35    |
| 36    |
| 37    |
| 38    |
| 170   |

#### Paper Task

Choose the best measure:

- Mean
- Median
- Mode

Write one sentence explaining your answer.

#### Excel Task

Calculate:

- Mean
- Median
- Mode

Write all three formulas.

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★ **New Topic – Range**

**Q6. Calculate the Range**

| Value |
|-------|
| 12    |
| 18    |
| 25    |
| 31    |
| 40    |

**Paper Task**

- Calculate the Range.

**Excel Task**

- Calculate the Range.
  - Write the Excel formula.
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### Q7. Calculate the Range

| Value |
|-------|
| 48    |
| 29    |
| 55    |
| 37    |
| 41    |

#### Paper Task

- Calculate the Range.

#### Excel Task

- Calculate the Range.
  - Write the Excel formula.
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### Q8. Interpretation

Two classes scored:

#### Class A

| Value |
|-------|
| 20    |
| 22    |
| 24    |
| 26    |
| 28    |

#### Class B

| Value |
|-------|
| 10    |
| 20    |
| 30    |
| 40    |
| 50    |

#### Paper Task

1. Calculate the **Range** for both classes.
2. Which class has **more spread**?

#### Excel Task

Calculate the Range for both classes using Excel.

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### Q9. Real Data Analyst Scenario

A manager wants to know **how much sales vary** between the lowest and highest selling stores.

Which statistical measure is most appropriate?

- A. Mean
- B. Median
- C. Mode
- D. Range

Explain your answer in one sentence.

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### Q10. Excel Formula Challenge

Write the Excel formulas for:

1. Mean
2. Median
3. Mode
4. Range

Use the range **B2:B6**.

**Example:**

=AVERAGE(B2:B6)

=MEDIAN(B2:B6)

=MODE.SNGL(B2:B6)

=MAX(B2:B6)-MIN(B2:B6)

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### ❖ Excel Functions Covered

| Concept | Excel Function         |
|---------|------------------------|
| Mean    | =AVERAGE()             |
| Median  | =MEDIAN()              |
| Mode    | =MODE.SNGL()           |
| Range   | =MAX(range)-MIN(range) |

**New Concept Learned in S4: Range** — the difference between the highest and lowest value in a dataset. In later sets, we'll build on this by introducing **Variance**, then **Standard Deviation**, while continuing to mix earlier topics so you retain them.